Credit Card Fraud Detection Using Machine Learning

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Project Type

Supervised Machine Learning

Project Overview

This project develops a supervised machine learning model to detect fraudulent credit card transactions. Since fraud cases are extremely rare, special techniques were used to handle class imbalance and improve fraud detection.

Problem Statement

Automatically classify transactions as legitimate or fraudulent while reducing financial losses and unnecessary investigations.

Objectives

Perform preprocessing, EDA, feature engineering, imbalance handling, model training, hyperparameter tuning, threshold optimization, business analysis and deployment.

Dataset

The dataset contains Time, Amount, PCA-transformed features V1 to V28, and the target column Class.

Preprocessing

Checked missing values, removed duplicates, created a stratified sample, engineered Amount_log and Hour features, and applied scaling where required.

EDA

Visualized class distribution, amount distribution, time distribution, boxplots and correlation heatmaps.

Class Imbalance

Compared Original Data, SMOTE Oversampling and Random Under Sampling.

Models

Built Logistic Regression, Random Forest and XGBoost models.

Hyperparameter Tuning

Used RandomizedSearchCV to find better XGBoost parameters.

Threshold Optimization

Optimized the classification threshold using the best F1-score and Recall ≥ 0.90 instead of always using 0.5.

Business Analysis

Calculated money saved, investigation cost, money lost and net benefit for different thresholds.

Deployment

Saved the final trained pipeline using Joblib as fraud_detection_model.pkl.

Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Imbalanced-learn, XGBoost, Joblib and Jupyter Notebook.

Conclusion

The tuned XGBoost model produced the best overall performance and was selected as the final deployment model.